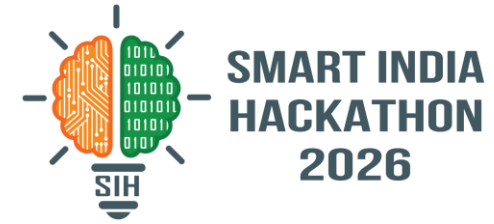
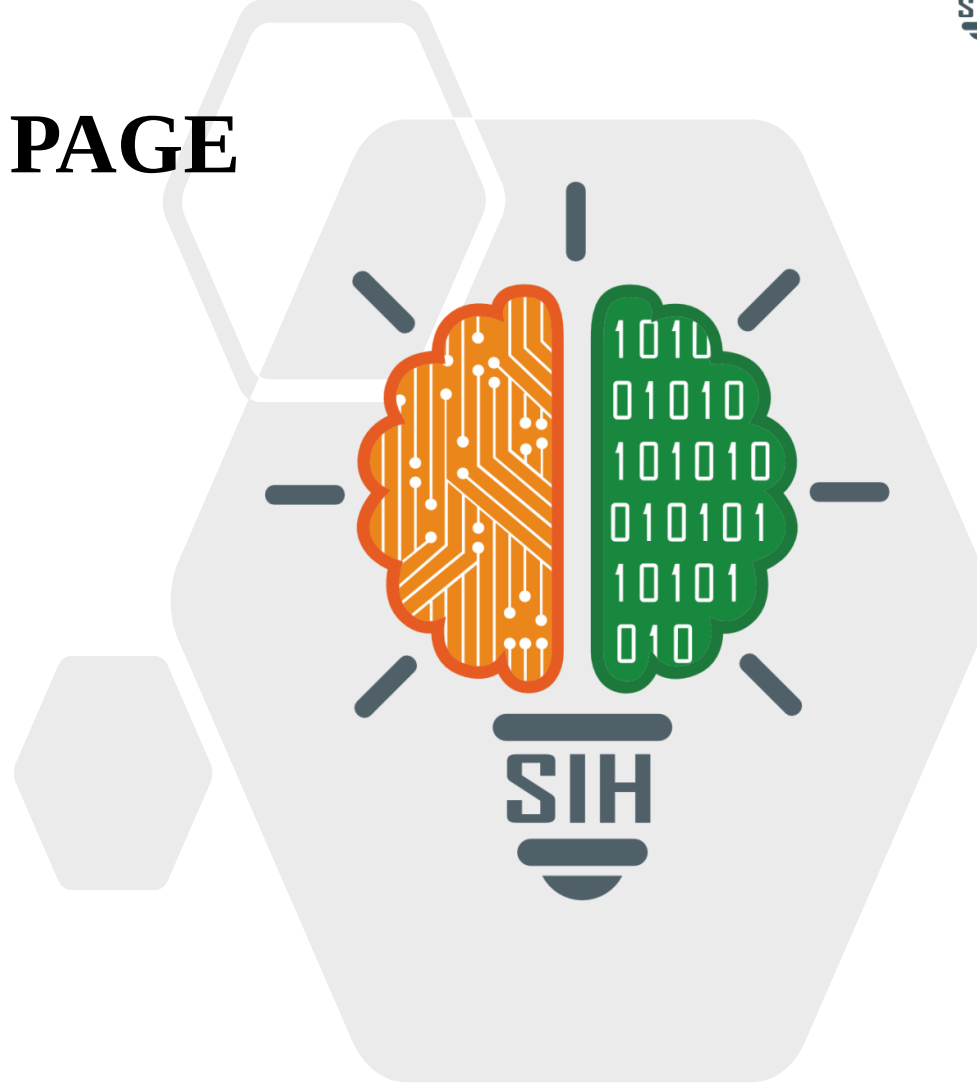


SMART INDIA HACKATHON 2026



TITLE PAGE

- Problem Statement ID – SIH26156
- Problem Statement Title- Universal Log
Pre-processing Framework
- Theme- Blockchain & Cybersecurity
- PS Category- Software
- Team ID - 120715
- Team Name- TEAM LOGUNIFYERS



ULPF - Universal Log Pre-processing Framework

(One Universal Event Language for Heterogeneous Security Logs)

Universal • Lossless • Traceable • Extensible • Air-Gapped

The Problem



Different vendors generate logs in **different formats and field names**



Source-specific **parsers** create high integration effort



Fragmented telemetry slows correlation, **SIEM** onboarding and analytics



FortiGate



Cisco ASA



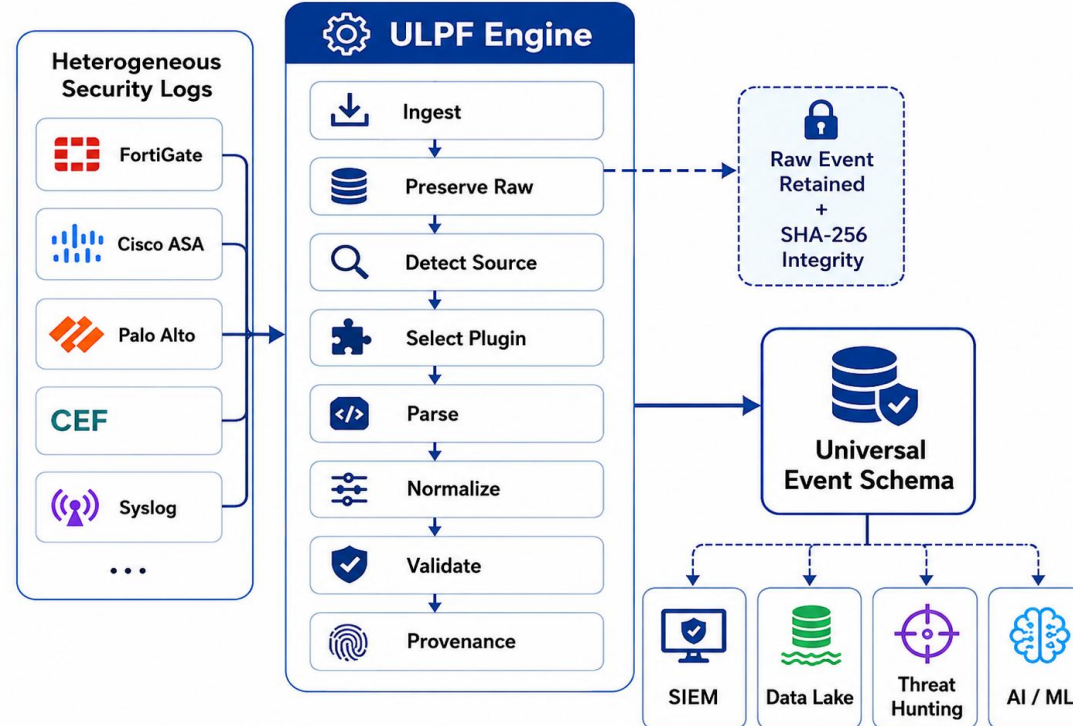
Palo Alto

CEF



Syslog

Our Proposed Solution



Innovation / Uniqueness



Lossless by Design

Original raw event is preserved with cryptographic integrity.



Field-Level Traceability

Every normalized field maps back to its original source.



Plug-and-Play Parser Plugins

Easily add or update parsers without engine changes.



Air-Gap Ready

Designed for secure, offline deployments in restricted environments.



Safe Unknown-Source Handling

Unknown sources are captured safely for analysis, not lost.



Different vendors speak different log languages.
ULPF gives them **one universal language** — without losing the original evidence.



1 TECHNOLOGY STACK



Python



FastAPI



Pydantic



SQLite



HTML/CSS/JS



Docker

REST API • JSON • NDJSON •
Offline-first architecture

3 RAW TO NORMALIZED

Raw:

```
src=ip=10.0.0.5
action=deny
protocol=..
...
```

Normalized:

```
source_ip = 10.0.0.5
event.action = DENY
network.protocol = TCP
...
```

+ Provenance

4 UNKNOWN - SOURCE PATH



Unknown
Log



Quarantine



Structure
Analysis



Plugin
Starter



Human
Review



Unknown logs are preserved and
reviewed — never silently guessed.

2 CORE PROCESSING PIPELINE

HETEROGENEOUS SECURITY LOGS



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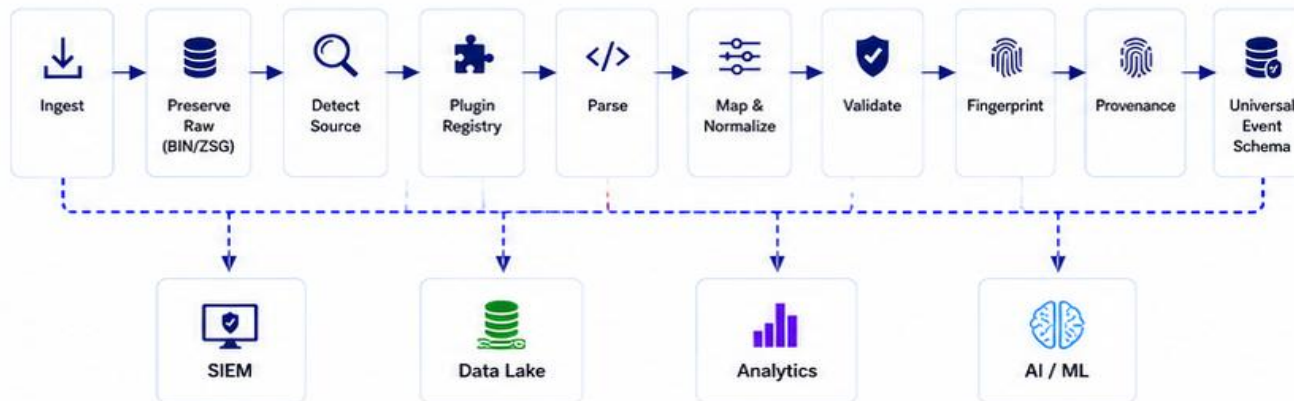
Palo Alto



CEF



Syslog



5 WORKING PROTOTYPE

VENDOR-AGNOSTIC • LOSSLESS • TRACEABLE

Ingestion

Paste or batch process logs

```
date=2026-08-26 time=17:08:00 tz="+0530" devname="FGT-EDGE-01" devid="FG20MT0000000001"
logid="0000000001" type="traffic" subtype="forward" level="notice" vs="root" srcip=10.0.0.5
srcport=52134 srcintf="port2" dstip=8.8.8.8 dstport=443 dstintf="wan1" srccountry="Reserved"
dstcountry="United States" sessionid=12345 proto=6 action="accept" policyid=7
service="HTTPS"trandip="noop" duration=3 sensbyte=982 rcvbyte=2048 msg="Allowed HTTPS session"
```

Process as single event Process non-empty lines Upload file

PROCESSING STATES

Latest batch result

STATUS	Device	Event ID	Action
STORED	fortigate	evt_5a109a5c3f4a4a2f4b0e7f9f4c	Inspect
STORED	cisco_asa	evt_11c0d77093a402072a0a4b1a677961	Inspect
STORED	palo_alto	evt_51d81e3a7e42c39e4b7f02086c711	Inspect
STORED	cef	evt_83938c824e74483a3a3c9c3f3a3d3f	Inspect
STORED	syslog	evt_7f4c7067a74a4a2f4b0e7f9f4c	Inspect
STORED	fortigate	evt_74866a18114959807c1172f4a4a407	Inspect
STORED	cisco_asa	evt_5d2ab3c8c3d4e4a2f4b0e7f9f4c	Inspect
STORED	palo_alto	evt_7077b0c87f4a4a2f4b0e7f9f4c	Inspect
STORED	cef	evt_f8dc6c8b1947b1a1a3e4b23515f4e4d	Inspect

HERO EVENT INSPECTOR

Fortinet ALLOW event
fortigate 1.0.0 mapping 1.0.0

RAW	PARSED	NORMALIZED
Exact original payload	Source-shaped attributes	Universal Event Schema
date=2026-08-26 time=17:08:00 tz="+0530" devname="FGT-EDGE-01" devid="FG20MT0000000001" logid="0000000001" type="traffic" subtype="forward" level="notice" vs="root" srcip=10.0.0.5 srcport=52134 srcintf="port2" dstip=8.8.8.8 dstport=443 dstintf="wan1" srccountry="Reserved" dstcountry="United States" sessionid=12345 proto=6 action="accept" policyid=7 service="HTTPS"trandip="noop" duration=3 sensbyte=982 rcvbyte=2048 msg="Allowed HTTPS session"	{ "action": "accept", "date": "2026-08-26", "devid": "FG20MT0000000001", "devname": "FGT-EDGE-01", "dstcountry": "United States", "dstintf": "wan1", "dstip": "8.8.8.8", "dstport": "443", "duration": "3", "level": "notice", "logid": "0000000001", "msg": "Allowed HTTPS session", "policyid": "7", "proto": "6", "rcvbyte": "2048", "sensbyte": "982", "service": "HTTPS", "sessionid": "12345", "srccountry": "Reserved", "srcintf": "port2", "srcip": "10.0.0.5", "srcport": "52134", "subtype": "forward", "time": "17:08:00", "type": "traffic" }	{ "@timestamp": "2026-08-26T17:08:00Z", "@version": 1, "log": { "type": "ALLOW", "category": "Network", "severity": "NOTICE", "type": "connection" }, "event": { "action": "ALLOW", "category": "Network", "severity": "NOTICE", "type": "connection" }, "event_id": "evt_5a109a5c3f4a4a2f4b0e7f9f4c", "timestamp": { "fortigate": { "devid": "FG20MT0000000001", "dstcountry": "United States", "dstintf": "wan1", "dstip": "8.8.8.8", "dstport": "443", "level": "notice", "logid": "0000000001", "msg": "Allowed HTTPS session", "policyid": "7", "rcvbyte": "2048", "sensbyte": "982", "sessionid": "12345", "time": "17:08:00" } } }

1 FEASIBLE TODAY

01 END-TO-END PROCESSING



Ingest → Preserve → Detect → Parse
→ Normalize → Validate → Trace

Complete preprocessing lifecycle implemented as one modular pipeline.

02 5 REPRESENTATIVE ADAPTERS



Prototype validates extensibility across five heterogeneous source types.

03 AIR-GAP READY CORE



Deterministic processing requires no external cloud or AI API for known log sources.

04 EASY TO INTEGRATE



ULPF complements the existing security stack — it does not replace it.

2 CHALLENGE → OUR STRATEGY

CHALLENGE

VENDOR DIVERSITY
Thousands of vendors, products & proprietary log structures

FORMAT EVOLUTION
Firmware updates may change field names or log syntax

UNKNOWN / MALFORMED LOGS
An unsupported source can produce unsafe or incorrect normalization

FORENSIC DATA LOSS
Normalization can remove vendor-specific or forensic information

HIGH EVENT VOLUME
Production environments generate extremely large event volumes

RESTRICTED NETWORKS
Air-gapped / offline environments with zero internet access

OUR STRATEGY

MODULAR PLUGIN ARCHITECTURE

Vendor-specific logic remains outside the core engine. New sources are added as independent parser plugins.

VERSIONED PARSERS + MAPPINGS

Parser and mapping versions keep transformations auditable and allow controlled updates without breaking the core.

QUARANTINE → ANALYZE → VALIDATE

Unknown telemetry is preserved and reviewed before mapping creation. No silent misclassification.

RAW-FIRST PRESERVATION

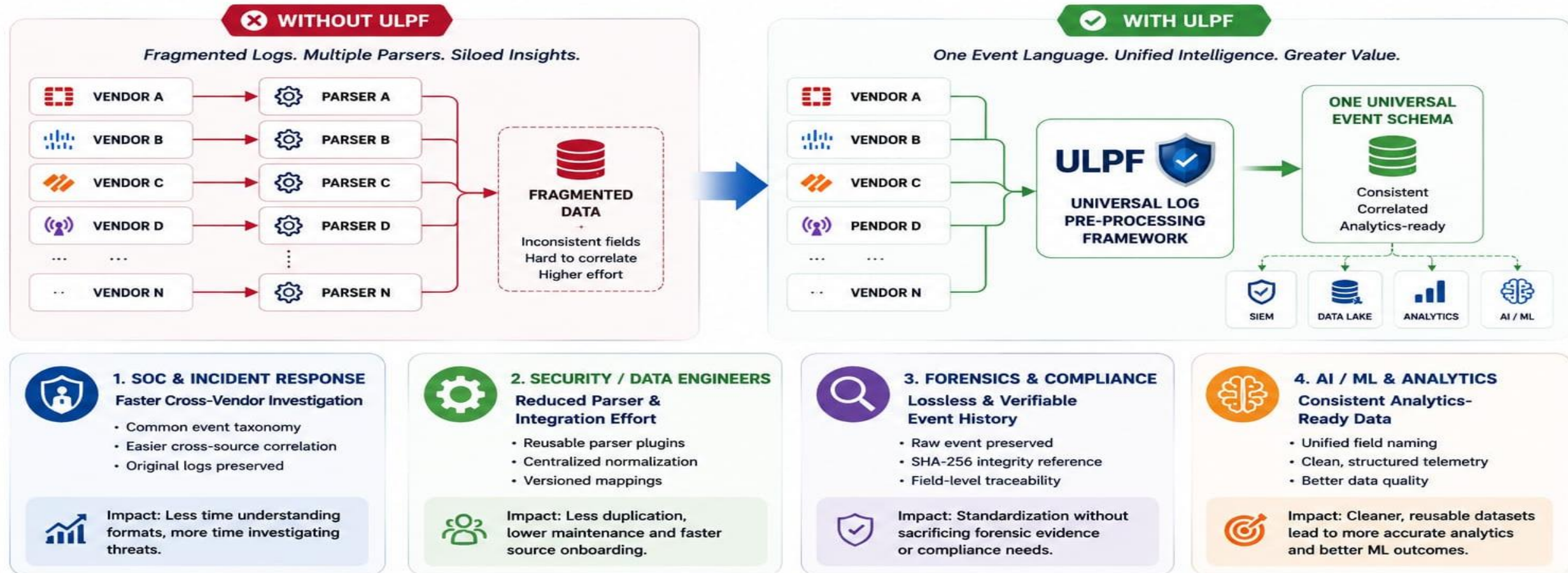
Raw event + SHA-256 integrity + Provenance ensures lossless and verifiable transformation.

EVENT-STREAM + HORIZONTAL SCALE

Queue-based ingestion with stateless workers enables horizontal scale-out for high throughput.

OFFLINE + CONTAINERIZED

Fully containerized local deployment with no mandatory external dependencies.



NORMALIZE ONCE • USE EVERYWHERE • PRESERVE EVERYTHING

ULPF converts fragmented vendor telemetry into reusable security intelligence without sacrificing original evidence.

1 REQUIREMENT FOUNDATION



SIH 2026 / NTRO Problem Statement 26156

Key requirements studied:

- ✓ Lossless raw-event preservation
- ✓ Common event taxonomy
- ✓ Raw ↔ normalized traceability
- ✓ Plug-and-play onboarding
- ✓ SIEM / Data Lake interoperability
- ✓ AI/ML-ready telemetry
- ✓ Air-gapped deployment



NTRO
Universal Log
Pre-processing
Framework (ULPF)

2 STANDARDS & LOGGING

NIST

NIST SP 800-92
Guide to Computer
Security Log Management

Studied for:

- Enterprise log management
- Security-event collection
- Reliable retention
- Incident investigation

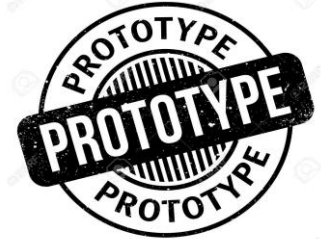
IETF

IETF RFC 5424
The Syslog Protocol

Studied for:

- Syslog message structure
- Structured data
- Timestamp & message formatting
- Vendor extensions

QUICK ACCESS



 **GitHub**

3 CYBERSECURITY SCHEMAS



OCSF
Open Cybersecurity
Schema Framework

- Vendor-neutral taxonomy
- Event classes & categories
- Extensible security schema



ECS
Elastic Common
Schema

- Common field naming
- Standardized data types
- Cross-source correlation
- Reusable analytics

4 REAL-WORLD SOURCE SPECIFICATIONS



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Palo Alto
PAN-OS



Common Event
Format



RFC 5424
Syslog



Used to develop source detection, deterministic parsers
and field mappings.

RESEARCH INSIGHT

“ Schemas define the common language.
ULPF builds the translation pipeline that gets
heterogeneous raw telemetry into that language —
losslessly and traceably. ”



Standards
Informed



Real-World
Validated



Secure &
Traceable



Analytics
Ready